

Practice Worksheet 3: Pseudocode

CS 101 Algorithmic Problem Solving

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1. Trick or Treat

Miles and Gwen have dressed up as Spiderman and Spiderwoman for Halloween and gone trick or treating.

After a long night, they come back home and sit down to start counting their earned sweets. Miles, just by looking over at Gwen's basket, realizes that they both didn't get the same amount of candy. The two collectively decide that they will sum their candies up and split them equally.

Given values of X and Y , the number of candies Miles and Gwen got respectively, determine if it is possible to split the candies such that both of them get an equal amount.

Constraints

- $X, Y \in \mathbb{N}$
- $1 \leq X, Y \leq 10^5$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and Y respectively.

The output must be "YES" if an equal distribution is possible and "NO" otherwise.

Sample

Input	Output
8 6	YES
10 11	NO

In the first case, $(X, Y) = (8, 6)$. Miles got 8 candies and Gwen got 6; these can be equally divided into 7 candies each.

In the second case, $(X, Y) = (10, 11)$. Miles got 10 candies and Gwen got 11; these cannot be equally divided.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
5 5	YES
10^5 1	NO
123 12	NO

Pseudocode

```
Input: X,Y
Output:
if (X+Y)%2==0:
    print("YES")
else:
    print("NO")
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 123, 12

```
if (123+12)%2==0: (false)
    print("YES")
else:
    print("NO")
```

Output: NO

Input: 5, 5

```
if (5+5)%2==0: (true)
    print("YES")
else:
    print("NO")
```

Output: YES

2. Football Match

Chef is watching a football match. The current score is $A : B$, that is, team 1 has scored A goals and team 2 has scored B goals. Chef wonders if it is possible for the score to become $C : D$ at a later point in the game (i.e. team 1 has scored C goals and team 2 has scored D goals).

Can you help Chef by answering his question?

Constraints

- $0 \leq A, B, C, D \leq 100$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of A, B, C and D respectively.

The output shall state if the given score transitioning is Possible or Impossible.

Sample

Input	Output
1 5 3 5	Possible
3 4 2 6	Impossible

In the first case, The current score is 1:5. If team 1 scores 2 more goals, the score will become 3:5. Thus 3:5 is a possible score.

In the second case, The current score is 3:4. It can be proven that no non-negative pair of integers (x, y) exists such that if team 1 scores x more goals and team 2 scores y more goals the score becomes 2:6 from 3:4. Thus in this case 2:6 is an impossible score.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2 2 2 2	Possible
0 1 8 0	Impossible
9 9 7 13	Impossible

Pseudocode

```

Input: A, B, C, D
Output:
if A<=C and B<=D:
    print("Possible")
else:
    print("Impossible")

```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 2, 2, 2, 2

```

if 2<=2 and 2<=2: (true)
    print("Possible")
else:
    print("Impossible")

```

Output= Possible

Input: 0, 1, 8, 0

```

if 0<=8 and 1<=0: (false)
    print("Possible")
else:
    print("Impossible")

```

Output= Impossible

3. Vacation

You finally get the time to go on a vacation after a tough semester. You have planned for two trips during this vacation—one with your family and the second with your friends.

The family trip will take X days and the trip with friends will take Y days. Currently, the dates are not decided but the vacation will last only for Z days.

You can only be in at most one trip at any time and once a trip is started, you must complete it before the vacation ends. Planning and packing are already done and will not take time.

Given X, Y , and Z , you want to see if you will be able to go on both the trips?

Constraints

- X, Y, Z
- $1 \leq X, Y, Z \leq 1000$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of X , Y and Z respectively.

The output must contain a single string—"YES" if you can go on both the trips and "NO" if not.

Sample

Input	Output
1 2 3	YES
2 2 3	NO

In the first case, $(X, Y, Z) = (1, 2, 3)$. The total duration of the trips is $1 + 2 = 3$ days which fits within the vacation days. You can go on both the trips.

In the second case, $(X, Y, Z) = (2, 2, 3)$. The total duration of the trips is $2 + 2 = 4$ days which exceeds the vacation days. You cannot go on both the trips.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
3 6 12	YES
2 2 4	YES
3 8 9	NO

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
4 5 8	NO
1 4 7	YES
1 1 4	YES

Pseudocode

```

Input: X, Y, Z
Output:
ans= X+Y
if ans<=Z:
    print("YES")
else:
    print("NO")

```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 2, 2, 4

```
ans= 2+2
if ans<=4: (true)
    print("YES")
else:
    print("NO")
```

Output= YES

Input: 3, 6, 12

```
ans= 3+6
if ans<=12: (true)
    print("YES")
else:
    print("NO")
```

Output= YES

4. Enrolment Time

For the upcoming fall semester, the Registrar's Office has decided to keep a total of X seats for the CS113 course. Upon conducting a student interest survey, it was found that Y students are interested in enrolling into CS113.

Given values of X and Y , determine the minimum number of extra seats that have to be added so that every interested student gets a seat.

Constraints

- $X, Y \in \mathbb{N}$
- $1 \leq X, Y \leq 10^5$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and Y respectively.

The output must be a single integer denoting the number of extra seats that must be added.

Sample

Input	Output
24 27	3
13 13	0

In the first case, $(X, Y) = (24, 27)$. 24 seats are currently allotted to CS113 and 27 students are interested in taking up the course, so 3 more seats need to be added.

In the second case, $(X, Y) = (13, 13)$. 13 seats are currently allotted to CS113 and 13 students are interested in taking up the course, so no more seats need to be added.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
40 57	17
25 23	0
31 62	31

Pseudocode

```
Input: X, Y
Output:
if Y<=X:
    print(0)
else:
    print(Y-X)
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 25, 23

```
if 23<=25: (true)
    print(0)
else:
    print(Y-X)
```

Output = 0

Input: 31, 62

```
if 62<=31: (false)
    print(0)
else:
    print(Y-X)
```

Output = 31

5. Taking a flight

Amna is getting on a flight and has 3 bags with her. She will submit 2 of them for check-in, and will take 1 to the airplane cabin with her as carry-on. The bags weigh A, B, and C kg each. The airline restricts, for a single passenger, the sum of the weights of the checked-in bags to no more than D kg and of the carry-on bags to no more than E kg. In case of excess, the passenger has to pay a fine.

Given A, B, C, D, and E, Find out if Amna can take her bags with her without having to pay a fine.

Constraints

- A, B, C, D, E
- $1 \leq A, B, C \leq 10$
- $15 \leq D \leq 20$
- $5 \leq E \leq 10$

Interaction

The input comprises a single line containing 5 space separated integers denoting the values of A, B, C, D and E respectively.

The output should contain a single word, YES, if Amna can take the flight without paying a fine, and NO if she will have to pay a fine.

Sample

Input	Output
1 1 1 15 5	YES
8 7 6 15 5	NO

In the first case, $(A, B, C, D, E) = (1, 1, 1, 15, 5)$, Amna can check in the first and second bags as their sum is less than 15 kg and carry the third bag as it weighs less than 5 kg. The output is YES.

In the second case, $(A, B, C, D, E) = (8, 7, 6, 15, 5)$. No combination of bags meets any of the limits. The output is NO.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
8 5 7 15 6	YES
8 8 9 16 5	NO
10 10 10 20 10	YES

Pseudocode

```

Input: A, B, C, D, E
Output:
W1=A+B
W2=B+C
W3=C+A
if W1<=D and C<=E:
    print("YES")
elif W2<=D and A<=E:
    print("YES")
elif W3<=D and B<=E:
    print("YES")
else:
    print("NO")

```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 8 5 7 15 6

```

W1=8+5
W2=5+7
W3=7+8
if W1<=15 and C<=6: (false)
    print("YES")
elif W2<=15 and A<=6: (false)
    print("YES")
elif W3<=15 and B<=6: (true)
    print("YES")
else:
    print("NO")
Output = YES

```

Input: 8 8 9 16 5

```

W1=8+8
W2=8+9
W3=9+8
if W1<=16 and C<=5: (false)
    print("YES")
elif W2<=16 and A<=5: (false)
    print("YES")
elif W3<=16 and B<=5: (false)
    print("YES")
else:
    print("NO")
Output= NO

```

6. Harley and the Chocolate Factory

Notorious chocolatier, Billy Bonka, has hidden a golden ticket in some of his chocolates. These ticket containing packets are incredibly rare and rumors say that there's only a dozen

such packets around the globe. Bonka has announced that anyone who can get their hands on one of these precious tickets will be invited to visit his state-of-the-art chocolate factory.

Harley, who has been a fan of Bonka's chocolates since she was young, is infatuated by the idea of visiting Bonka's factory. She breaks open her piggy bank and finds that she has X 5-cent coins and Y 10-cent coins. Harley knows a packet of chocolate costs Z cents and wants to know how many packets she can purchase.

Given values of X , Y , and Z , determine how many packets Harley can purchase.

Constraints

- $X, Y, Z \in \mathbb{Z}$
- $1 \leq X, Y, Z \leq 1000$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of X , Y , and Z respectively.

The output must contain a single number denoting the number of chocolates Harley can afford.

Sample

Input	Output
10 10 10	15
3 1 8	3

In the first case, $(X, Y, Z) = (10, 10, 10)$. Harley has 10 5-cent coins and 10 10-cent coins. She has a total of 150 cents. Since each packet costs 10 cents, she can afford 15 packets.

In the second case, $(X, Y, Z) = (3, 1, 8)$. Harley has 3 5-cent coins and 1 10-cent coins. She has a total of 25 cents. Since each packet costs 8 cents, she can afford 3 packets.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4 4 1000	0
10 5 1	100
13 31 35	10

Pseudocode

Input: X, Y, Z

Output:

$A = (X * 5) + (Y * 10)$

$B = A // Z$

print(B)

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 4 4 1000

$A = (4 \cdot 5) + (4 \cdot 10)$
 $B = 60 // 1000$
 print(B)

Output = 0

Input: 10 5 1

Output:
 $A = (10 \cdot 5) + (5 \cdot 10)$
 $B = 100 // 1$
 print(B)

Output = 100

7. Make Uncle Proud

Uncle Roger prepared two dishes yesterday and assigned tastiness T_1 and T_2 to the first and second dish respectively. T_1 and T_2 can be any integer between 0 and N (both inclusive).

Unfortunately, he seems to have forgotten what tastiness he assigned to each dish and only remembers what the total tastiness S was ($S = T_1 + T_2$).

As there is no way to figure out the original assigned tastiness now, Uncle Roger at least wants to figure out what was the maximum difference in the tastiness of the two dishes.

Given values of N and S , determine the maximum difference between the two of Uncle Roger's dishes.

Constraints

- $N, S \in \mathbb{N}$
- $1 \leq N \leq 10^5$
- $1 \leq S \leq 2 \cdot 10^5$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of N and S respectively.

The output must be a single integer denoting the maximum difference between the tastiness of the two dishes.

Sample

Input	Output
3 1	1
2 3	1

In the first case, $(N, S) = (3, 1)$. The maximum tastiness for each dish could be 3 and the sum of the tastiness for each dish is 1. Therefore, the cases where dishes have maximum difference are $(T_1, T_2) = (0, 1)$ and $(T_1, T_2) = (1, 0)$ where the maximum difference is 1.

In the second case, $(N, S) = (2, 3)$. The maximum tastiness for each dish could be 2 and the sum of the tastiness for each dish is 3. Therefore, the cases where dishes have maximum difference are $(T_1, T_2) = (2, 1)$ and $(T_1, T_2) = (1, 2)$ where the maximum difference is 1.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4 4	4
1356 2134	578
$10^3 \ 2 \cdot 10^3$	0

Pseudocode

Input: N, S

Output:

T1=abs(S-N)

ans=abs(N-T1)

print(ans)

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 1000, 2000

Input: 1356, 2134

T1=abs(2000-1000)
ans=abs(1000-1000)
print(ans)

T1=abs(2134-1356)
ans=abs(1356-778)
print(ans)

Output = 0

Output = 578

8. Tax Money

In Chefland, a tax of rupees 10 is deducted if the total income is strictly greater than rupees 100.

Given that total income is X rupees, find out how much money you get.

Constraints

- $X \in \mathbb{N}$
- $1 \leq X \leq 1000$

Interaction

The input comprises a single line containing an integer denoting the value of X .

The output must contain a single number denoting the amount of money you get.

Sample

Input	Output
5	5
105	95

In the first case, Your total income is 5 rupees which is less than 100 rupees. Thus, no tax would be deducted and you get 5 rupees.

In the second case, Your total income is 105 rupees which is greater than 100 rupees. Thus, a tax of 10 rupees would be deducted and you get $105 - 10 = 95$ rupees.

Proposed Solution

```
new_income = 0
if X < 100:
    new_income = x - 10
print(new_income)
```

Dry Run

Using any two of the inputs provided in the Sample section above, dry run the proposed pseudocode in the space below.

```
Input: X
new_income = 0
X=5
if 5 < 100: (true)
    new_income = 5-10
print(new_income)
```

Output = -5

```
Input: X
new_income = 0
X=105
if 105 < 100: (false)
    new_income = 105-10
print(new_income)
```

Output = 0 (no else statement)

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

the sign in second line should be ">" (greater than) instead of "<" less than. The variable in line 3 should be capital "X"

there is no else statement, the program will show zero if condition is false

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

```
Input: X
Output:
new_income = 0
if X > 100:
    new_income = X-10
    print(new_income)
else:
    print(X)
```

9. Volume Control

Peter is watching TV. The current volume of the TV is X . Pressing the “volume up” button increases the volume by 1 while pressing the “volume down” button decreases the volume by 1. Peter wants to change the volume from X to Y .

Given values of X and Y , find the number of button presses it takes to change the volume from X to Y .

Constraints

- $X, Y \in \mathbb{N}$
- $1 \leq X, Y \leq 100$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and Y respectively.

The output must contain a single number denoting the minimum number of button presses to change volume from X to Y .

Sample

Input	Output
50 54	4
100 60	40

In the first case, $(X, Y) = (50, 54)$. Peter wants to change the volume from 50 to 54 and this can be done by 4 presses of the “volume up” button.

In the second case, $(X, Y) = (100, 60)$. Peter wants to change the volume from 100 to 60 and this can be done by 40 presses of the “volume down” button.

Proposed Solution

```
print(X-Y)
```

Dry Run

Using the inputs provided in the Sample section above, dry run the proposed code solution below.

Input: 50, 54

Input: 100, 60

`print(50-54)`

`print(100-60)`

Output = -4 (expected=4)

Output = 40

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number(s) and the error(s) in each line.

The code only works fine if $X > Y$, but for smaller values of X , the value printed will be in negative integers.

Correct Solution

Rewrite the line(s) of code that you mentioned above with their errors corrected.

Input: X, Y

Output:

A=abs(X-Y)

print(A)

10. Introduction to Hackerrank

The guide to login/sign-up on Hackerrank tests is available here.

Problems are available on this link as a short introduction to programming:

<https://hr.gs/cs101-pw3-fall24>

Done in lab of Sept 3rd

Rough Work